

Fig. 5: Output with both first and second-highest priority

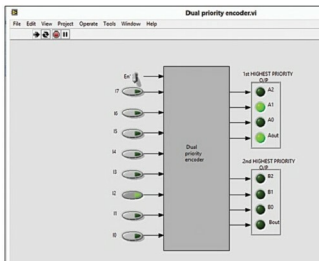


Fig. 6: Output with only first-highest priority

Select functions as shown in Fig. 2, except first Select function of I7.

Step 23. Connect output of Select function corresponding to I1 to the 'f' input of the Select function corresponding to I2. Similarly, repeat the same connection at the Select functions corresponding to I3 through I6 inputs, except for I7. And then create a constant 0 at 'f' input of Select function corresponding to input I0.

Step 24. Select another Case Structure and drop it on to Block Diagram inside the main Case Structure. If labels appear as True and False, modify them to 0 and 1, respectively and then add six more cases and label them as 2, 3, 4, 5, 6 and 7.

Step 25. Connect the output of the

second Select function of I7 to selector terminal '?' of the inner lower Case Structure. And then repeat steps 18 and 19 for outputs B2, B1 and B0.

Step 26. Select Programming-> Numeric-> Compound Arithmetic and drop it on to Block Diagram. Extend it so that it has eight inputs. Now, connect the eight outputs from Boolean (0, 1) functions to the eight inputs of Compound Arithmetic. Change the mode of Compound Arithmetic to 'Add' by right clicking on it and select Change Mode-> Add option. This will add a '+' sign to it to make it an adder function.

Step 27. Select Programming-> Comparison-> Greater? and drop it on to Block Diagram inside the main Case Structure.

Give the output of Compound Arithmetic adder to the first input 'x' of Greater?. Place cursor on the second input 'y' and right clicking on it create a constant of value 1.

Step 28. Connect the output of Greater? to Round LED BVALID and draw another wire from output of Compound Arithmetic adder to input of another comparison function 'Greater Than 0?'. Give the output of it to another Round LED AVALID.

Step 29. Press Ctrl+b keys to remove broken wires and press Ctrl+u keys to clean up the diagram on the Block Diagram as shown in Fig. 3.

Step 30. Change the case to False condition in main Case Structure and

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

connect all the Tunnels on right side of Case Structure to False (F) constant. Then all the Tunnels will be closed as shown in Fig. 4.

Step 31. Return to Front Panel. Save the VI and press the Run Continuously button to check for the operation of dual-priority encoder. Give different inputs and check the outputs for identifying the first-highest and second-highest priority request.

Checking the output

Fig. 5 shows the result for dual-priority encoder operations for first-highest and second-highest priority requests when the inputs I7, I5, I3 and I0 are asserted. Here, I7 has the highest priority, after that the next-highest priority will be given to I5. Hence, all the output LEDs A2, A1 and A0 glow, which is equivalent to 111 (also a binary representation of 7) in the highest-priority box. In the second-highest-priority box, B2 and B0 are on and B1 is off, which is equivalent to 101 (also a binary representation of 5). The output LEDs both AVALID and BVALID are asserted, because more than one input is asserted.

Fig. 6 shows the result when only one input I2 is asserted. Here, output LEDs A2, A1 and A0 show the highest-priority request 2 as 010 (binary representation of 2) and the output LEDs B2, B1 and B0 are turned off. The output LED AVALID only is asserted, because we asserted only single input and there is no second-highest-priority request. Here, note that the circuit will be on only when the enable input (En) is in off condition, because it is an active low enable input. ●

The authors Paluru Sunitha and Naga Sunil Manohar Bole were third-year B.Tech (electronics and communication) students from Maharaj Vijayaraj Gajapathi Raj College of Engineering, Vizianagaram when they sent this project, under the guidance of Associate Professor M. Satyanarayana, last year